MAE3405 Assignment 3: Gas Turbine Design

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1. Introduction

We set out to examine the design point cruise calculations for a passenger aircraft with two high-bypass turbofan engines, the Pratt & Whitney PW4000 [1]. The particular configuration of this engine involves a twin spool system and separate hot and cold nozzles. The low pressure turbine drives the fan and the low pressure compressor, while the high pressure turbine drives the high pressure compressor. The engine uses Jet-A fuel. In the first part, we will determine the net thrust and TSFC from sea level to 15 km at Mach 0.75 and a fuel-air ratio of 3%. Then in the next part we will determine the maximum Mach number for each altitude where positive net thrust is achieved, at a fuel-air ratio of 2.5%. We will also limit the flight Mach number to 1. All ambient conditions are sourced from the International Standard Atmosphere table in the MAE3405 aide memoire [2], spaced at 0.5 km. All calculations and figures were sourced from a single Python script.

2. Part (a): Net thrust and TSFC

The schematic for each engine is shown in Figure 1 where each section of the engine is labelled alongside the different thermodynamic states.

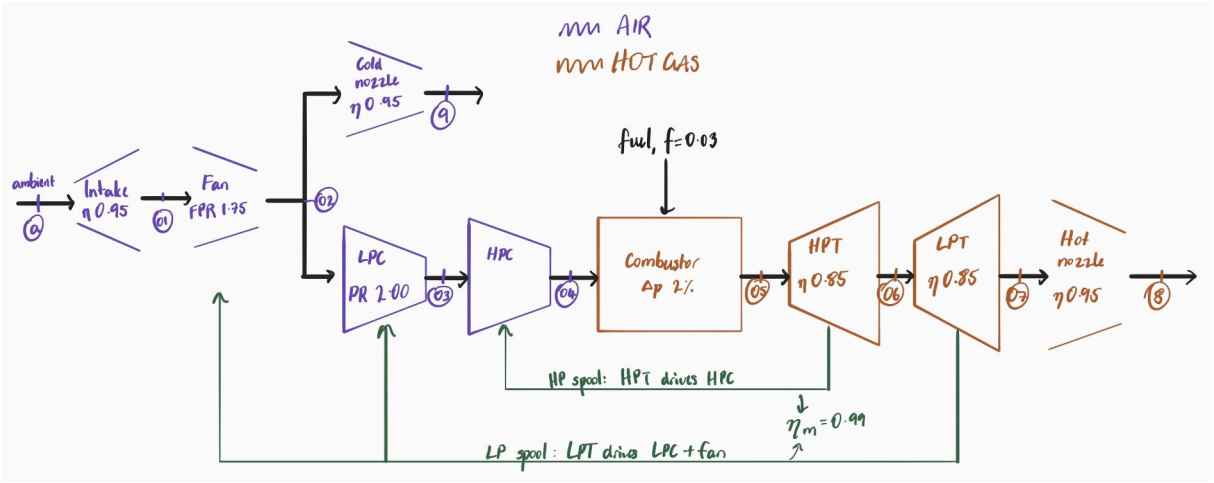


Figure 1: Labelled schematic of a single engine showing intake, fan, low pressure compressor, high pressure compressor, combustor, high pressure turbine, low pressure turbine, hot nozzle, and cold nozzle. Both spools and the bypass split are marked. Station numbering is consistent with written calculations and code.

The station numbering is consistent and is as follows: a is ambient, 01 is the intake exit, 02 is the fan exit, 03 is the low pressure compressor exit, 04 is the high pressure compressor exit, 05 is the combustor exit, 06 is the high pressure turbine exit, 07 is the low pressure turbine exit, 8 is the hot nozzle exit and 9 is the cold nozzle exit. Stations 01 to 07 are stagnation states, and a, 8 and 9 are static states.

The flow splits at station 02. The bypass stream goes straight to the cold nozzle (02 to 9), and the core stream continues through the two compressors. Figure 2 and Figure 3 are the P-v and T-s diagrams respectively showing the states and processes. They also show the constant pressure line. Neither figure is drawn to scale, but reveals a pattern that will govern our expected results.

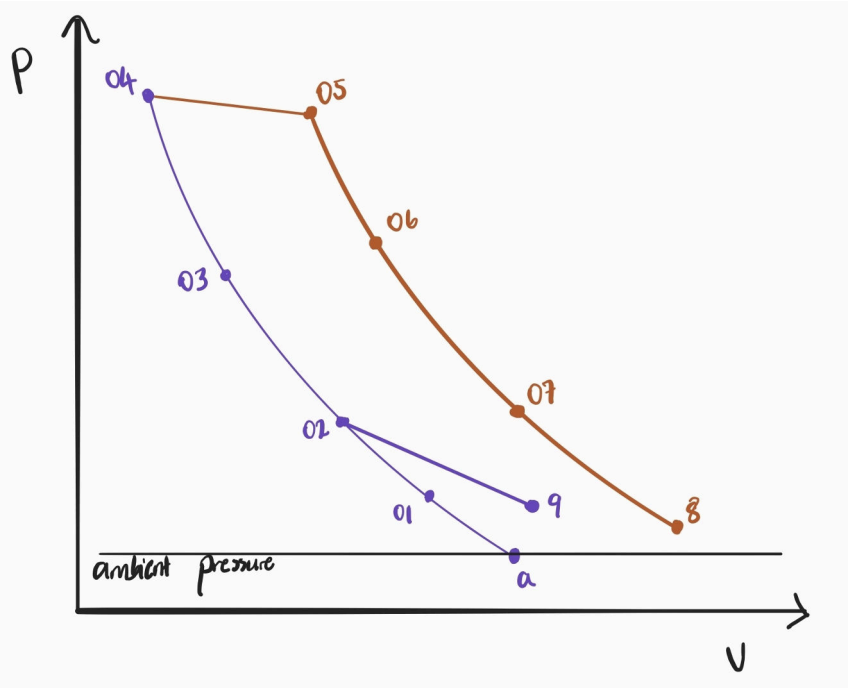


Figure 2: P-V diagram showing the different states and processes inside the engine. States and processes are not to scale.

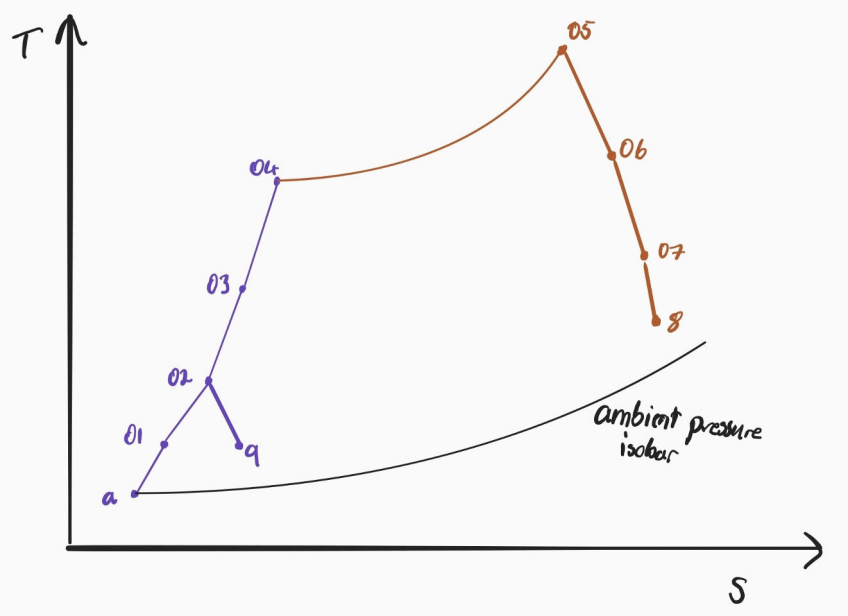


Figure 3: T-S diagram showing the different states and processes inside the engine. States and processes are not to scale.

2.1. Thermodynamic Calculations

We will assume the flow is steady and one-dimensional, and the working fluid is an ideal gas with constant properties. The cold values ($c_{pa} = 1005$ J/(kg K), $\gamma_a = 1.4$) apply up to the combustor inlet and in the bypass stream, and the hot values ($c_{pg} = 1148$ J/(kg K), $\gamma_g = 1.333$) apply from the combustor exit onwards. $R = 287$ J/(kg K) is used for both streams. These values, and all engine specifications, are taken from the assignment brief [1].

The intake captures a stream tube with the same area as the intake, $A = 5.067$ m^2, at the flight speed, so $\dot{m} = \rho_a C_a A$. The overall pressure ratio of 32 is the product of the fan, low pressure

compressor and high pressure compressor ratios. The bypass air is compressed by the fan only. The lower heating value of Jet-A is taken as 43.1 MJ/kg, the value listed for Jet A1, with a reference temperature of 298 K [2].

With these assumptions we can begin the calculations. For each state I will derive the general equations and then provide sample calculations at 11 km.

2.1.1. a → 01 Intake

The intake will be considered to be adiabatic, with a given isentropic efficiency of 0.95. The International Standard Atmosphere 1976 (ISA) table in the MAE3405 Aide Memoire [2] provides the ambient state. At 11 km, we have $T_a = 216.65$ K, $p_a = 22.63$ kPa, $\rho_a = 0.3639$ kg/m^3, and $a = 295.07$ m/s.

We can first calculate the stagnation temperature at 01, knowing that stagnation temperature is constant over an adiabatic process, using the stagnation temperature equation [2]:

$$T_{01} = T_{0a} = T_a + \frac{C_a^2}{2c_{pa}}, \tag{1}$$

with the speed $C_a = M \cdot a$. We can do the same for the stagnation pressure using the isentropic pressure relation modified by the isentropic efficiency,

$$\frac{p_{01}}{p_a} = \left[1 + \eta_i \frac{C_a^2}{2c_{pa}T_a}\right]^{\frac{\gamma_a}{\gamma_a - 1}}. \tag{2}$$

As an example, at 11 km,

$$T_{01} = 216.65 + \frac{(0.75 \cdot 295.07)^2}{2 \cdot 1005} = 241.02 \text{ K},$$

and with $C_a = 221.30$ m/s,

$$p_{01} = 22.63 \left[1 + \frac{0.95 \cdot 221.30^2}{2 \cdot 1005 \cdot 216.65}\right]^{\frac{1.4}{1.4-1}} = 32.29 \text{ kPa}.$$

2.1.2. Mass flows

The total air mass flow through the intake is set by the ambient density, the flight speed and the intake area,

$$\dot{m} = \rho_a C_a A, \quad A = \frac{\pi D^2}{4}, \tag{3}$$

where $D = 100$ in is the intake diameter. With a bypass ratio of $B = 4.9$, the flow splits at the fan into a core stream and a bypass stream,

$$\dot{m}_c = \frac{\dot{m}}{1+B}, \quad \dot{m}_b = \frac{B\dot{m}}{1+B}, \tag{4}$$

and the fuel is added to the core stream only, at the fuel-air ratio f,

$$\dot{m}_f = f\dot{m}_c. \tag{5}$$

As an example, at 11 km,

$$\dot{m} = 0.3639 \cdot 221.30 \cdot 5.067 = 408.06 \text{ kg/s},$$

$$\dot{m}_c = \frac{408.06}{1+4.9} = 69.16 \text{ kg/s}, \quad \dot{m}_b = \frac{4.9 \cdot 408.06}{1+4.9} = 338.90 \text{ kg/s},$$

$$\dot{m}_f = 0.03 \cdot 69.16 = 2.0749 \text{ kg/s}.$$

2.1.3. 01 → 04 Fan and compressors

The fan compresses the full air mass flow with a pressure ratio of $r_f = 1.75$, and the low pressure compressor acts on the core stream only, with a pressure ratio of $r_{\text{LPC}} = 2.0$. The high pressure compressor ratio is not given directly, but follows from the overall pressure ratio $r_o = 32$ being the product of the three compression ratios,

$$r_{\text{HPC}} = \frac{r_o}{r_f r_{\text{LPC}}}. \tag{6}$$

All three have an isentropic efficiency of $\eta_f = \eta_{\text{LPC}} = \eta_{\text{HPC}} = 0.85$. The exit stagnation pressure follows directly from the pressure ratio, the isentropic exit temperature follows from the isentropic relation, and the isentropic efficiency then gives the actual exit temperature. For the fan,

$$p_{02} = r_f p_{01}, \quad T_{02s} = T_{01}\left(\frac{p_{02}}{p_{01}}\right)^{\frac{\gamma_a - 1}{\gamma_a}}, \quad T_{02} = T_{01} + \frac{T_{02s} - T_{01}}{\eta_f}, \tag{7}$$

and the same three relations apply to the low pressure compressor (02 → 03) and the high pressure compressor (03 → 04).

As an example, at 11 km,

$$r_{\text{HPC}} = \frac{32}{1.75 \cdot 2.0} = 9.1429,$$

$$p_{02} = 1.75 \cdot 32.29 = 56.50 \text{ kPa}, \quad T_{02s} = 241.02 \cdot 1.75^{\frac{1.4-1}{1.4}} = 282.80 \text{ K},$$

$$T_{02} = 241.02 + \frac{282.80 - 241.02}{0.85} = 290.18 \text{ K},$$

$$p_{03} = 2.0 \cdot 56.50 = 113.00 \text{ kPa}, \quad T_{03s} = 290.18 \cdot 2.0^{\frac{1.4-1}{1.4}} = 353.73 \text{ K},$$

$$T_{03} = 290.18 + \frac{353.73 - 290.18}{0.85} = 364.95 \text{ K},$$

$$p_{04} = 9.1429 \cdot 113.00 = 1033.17 \text{ kPa}, \quad T_{04s} = 364.95 \cdot 9.1429^{\frac{1.4-1}{1.4}} = 686.79 \text{ K},$$

$$T_{04} = 364.95 + \frac{686.79 - 364.95}{0.85} = 743.58 \text{ K}.$$

2.1.4. 04 → 05 Combustor

The combustor has a stagnation pressure loss of 2%, so

$$p_{05} = (1 - 0.02) p_{04}. \tag{8}$$

The exit temperature follows from an energy balance per kilogram of core air, with enthalpies $h = c_p(T - T_{\text{ref}})$ [2] measured from the reference temperature $T_{\text{ref}} = 298$ K of the lower heating value $Q_{\text{LHV}} = 43.1$ MJ/kg. The enthalpy of the incoming air plus the heat released by the fuel,

reduced by the combustion efficiency $\eta_b = 0.98$, equals the enthalpy of the $(1 + f)$ kilograms of products,

$$c_{pa}(T_{04} - T_{\text{ref}}) + \eta_b f Q_{\text{LHV}} = (1 + f)c_{pg}(T_{05} - T_{\text{ref}}), \tag{9}$$

which rearranges to

$$T_{05} = T_{\text{ref}} + \frac{c_{pa}(T_{04} - T_{\text{ref}}) + \eta_b f Q_{\text{LHV}}}{(1 + f)c_{pg}}. \tag{10}$$

As an example, at 11 km,

$$p_{05} = (1 - 0.02) \cdot 1033.17 = 1012.51 \text{ kPa},$$

$$T_{05} = 298 + \frac{1005(743.58 - 298) + 0.98 \cdot 0.03 \cdot 43.1 \times 10^6}{(1 + 0.03) \cdot 1148} = 1748.35 \text{ K}.$$

2.1.5. 05 → 06 High pressure turbine

The high pressure turbine drives the high pressure compressor through the high pressure spool, which has a mechanical efficiency of $\eta_m = 0.99$. Per kilogram of core air, the work balance is

$$\eta_m(1 + f)c_{pg}(T_{05} - T_{06}) = c_{pa}(T_{04} - T_{03}), \tag{11}$$

so the exit temperature is

$$T_{06} = T_{05} - \frac{c_{pa}(T_{04} - T_{03})}{\eta_m(1 + f)c_{pg}}. \tag{12}$$

The turbine has an isentropic efficiency of $\eta_{\text{HPT}} = 0.85$, which gives the isentropic exit temperature,

$$T_{06s} = T_{05} - \frac{T_{05} - T_{06}}{\eta_{\text{HPT}}}, \tag{13}$$

and the exit stagnation pressure then follows from the isentropic relation with the hot gas properties,

$$p_{06} = p_{05}\left(\frac{T_{06s}}{T_{05}}\right)^{\frac{\gamma_g}{\gamma_g - 1}}. \tag{14}$$

As an example, at 11 km,

$$T_{06} = 1748.35 - \frac{1005(743.58 - 364.95)}{0.99(1 + 0.03) \cdot 1148} = 1423.28 \text{ K},$$

$$T_{06s} = 1748.35 - \frac{1748.35 - 1423.28}{0.85} = 1365.92 \text{ K},$$

$$p_{06} = 1012.51\left(\frac{1365.92}{1748.35}\right)^{\frac{1.333}{1.333 - 1}} = 376.93 \text{ kPa}.$$

2.1.6. 06 → 07 Low pressure turbine

The low pressure turbine drives both the low pressure compressor, which acts on the core stream, and the fan, which acts on the full flow of $(1 + B)$ kilograms per kilogram of core air. The work balance on the low pressure spool is therefore

$$\eta_m(1+f)c_{pg}(T_{06}-T_{07}) = c_{pa}[(T_{03}-T_{02})+(1+B)(T_{02}-T_{01})], \tag{15}$$

so the exit temperature is

$$T_{07} = T_{06} - \frac{c_{pa}[(T_{03}-T_{02})+(1+B)(T_{02}-T_{01})]}{\eta_m(1+f)c_{pg}}. \tag{16}$$

With an isentropic efficiency of $\eta_{\mathrm{LPT}} = 0.85$, the isentropic exit temperature and the exit stagnation pressure follow as for the high pressure turbine,

$$T_{07s} = T_{06} - \frac{T_{06}-T_{07}}{\eta_{\mathrm{LPT}}}, \quad p_{07} = p_{06}\left(\frac{T_{07s}}{T_{06}}\right)^{\frac{\gamma_g}{\gamma_g-1}}. \tag{17}$$

As an example, at 11 km,

$$T_{07} = 1423.28 - \frac{1005[(364.95-290.18)+(1+4.9)(290.18-241.02)]}{0.99(1+0.03)\cdot 1148} = 1110.07 \text{ K},$$

$$T_{07s} = 1423.28 - \frac{1423.28-1110.07}{0.85} = 1054.80 \text{ K},$$

$$p_{07} = 376.93\left(\frac{1054.80}{1423.28}\right)^{\frac{1.333}{1.333-1}} = 113.60 \text{ kPa}.$$

2.1.7. 07 → 8 and 02 → 9 Nozzles

Both nozzles are convergent with an isentropic efficiency of $\eta_j = 0.95$. The hot nozzle expands the core stream from the low pressure turbine exit (07) with the hot gas properties, and the cold nozzle expands the bypass stream from the fan exit (02) with the cold gas properties. The relations are written below for the hot nozzle. For the cold nozzle, 07 is replaced by 02, γ_g and c_{pg} by γ_a and c_{pa}, and the mass flow $\dot{m}_c(1+f)$ by $\dot{m}_b$.

We must first check whether each nozzle is choked. The critical pressure, at which the exit flow reaches Mach 1, is [2]

$$p_{8,c} = p_{07}\left[1 - \frac{1}{\eta_j}\left(\frac{\gamma_g-1}{\gamma_g+1}\right)\right]^{\frac{\gamma_g}{\gamma_g-1}}. \tag{18}$$

If $p_{8,c} > p_a$ the nozzle is choked, so the exit is sonic and the exit pressure is the critical pressure, $p_8 = p_{8,c}$. The exit temperature, velocity and density then follow, and the exit area follows from continuity,

$$T_8 = \frac{2T_{07}}{\gamma_g+1}, \quad C_8 = \sqrt{\gamma_g R T_8}, \quad \rho_8 = \frac{p_8}{RT_8}, \quad A_8 = \frac{\dot{m}_c(1+f)}{\rho_8 C_8}. \tag{19}$$

If instead $p_{8,c} \le p_a$, the nozzle is not choked and the flow expands fully to ambient, $p_8 = p_a$, so there is no pressure thrust. The isentropic efficiency then gives

$$T_{8s} = T_{07}\left(\frac{p_a}{p_{07}}\right)^{\frac{\gamma_g-1}{\gamma_g}}, \quad T_8 = T_{07} - \eta_j(T_{07}-T_{8s}), \quad C_8 = \sqrt{2c_{pg}(T_{07}-T_8)}. \tag{20}$$

As an example, at 11 km,

$$p_{8,c} = 113.60\left[1 - \frac{1}{0.95}\left(\frac{1.333-1}{1.333+1}\right)\right]^{\frac{1.333}{1.333-1}} = 59.20 \text{ kPa},$$

$$p_{9,c} = 56.50\left[1 - \frac{1}{0.95}\left(\frac{1.4-1}{1.4+1}\right)\right]^{\frac{1.4}{1.4-1}} = 28.76 \text{ kPa},$$

which are both greater than $p_a = 22.63$ kPa, so both nozzles are choked and

$$T_8 = \frac{2 \cdot 1110.07}{1.333+1} = 951.62 \text{ K}, \quad C_8 = \sqrt{1.333 \cdot 287 \cdot 951.62} = 603.38 \text{ m/s},$$

$$\rho_8 = \frac{59.20 \times 10^3}{287 \cdot 951.62} = 0.2168 \,\text{kg/m}^3, \quad A_8 = \frac{69.16(1+0.03)}{0.2168 \cdot 603.38} = 0.5447 \,\text{m}^2,$$

$$T_9 = \frac{2 \cdot 290.18}{1.4+1} = 241.81 \text{ K}, \quad C_9 = \sqrt{1.4 \cdot 287 \cdot 241.81} = 311.71 \text{ m/s},$$

$$\rho_9 = \frac{28.76 \times 10^3}{287 \cdot 241.81} = 0.4145 \,\text{kg/m}^3, \quad A_9 = \frac{338.90}{0.4145 \cdot 311.71} = 2.6233 \,\text{m}^2.$$

2.1.8. Net thrust and TSFC

The net thrust of each engine is the momentum flux leaving both nozzles, less the ram drag of the incoming air, plus the pressure thrust from each nozzle,

$$F = \dot{m}_c(1+f)C_8 + \dot{m}_b C_9 - \dot{m} C_a + (p_8 - p_a)A_8 + (p_9 - p_a)A_9. \tag{21}$$

The thrust specific fuel consumption is the fuel mass flow per unit of net thrust,

$$\text{TSFC} = \frac{\dot{m}_f}{F}. \tag{22}$$

As an example, at 11 km, the five terms of the net thrust are

$$\dot{m}_c(1+f)C_8 = 69.16(1+0.03) \cdot 603.38 = 42.98 \text{ kN},$$

$$\dot{m}_b C_9 = 338.90 \cdot 311.71 = 105.64 \text{ kN},$$

$$\dot{m} C_a = 408.06 \cdot 221.30 = 90.30 \text{ kN},$$

$$(p_8 - p_a)A_8 = (59.20 - 22.63) \times 10^3 \cdot 0.5447 = 19.92 \text{ kN},$$

$$(p_9 - p_a)A_9 = (28.76 - 22.63) \times 10^3 \cdot 2.6233 = 16.08 \text{ kN},$$

so the net thrust of each engine is

$$F = 42.98 + 105.64 - 90.30 + 19.92 + 16.08 = 94.3 \text{ kN},$$

or 189 kN for both engines. The thrust specific fuel consumption is then

$$\text{TSFC} = \frac{2.0749}{94.32 \times 10^3} = 2.20 \times 10^{-5} \text{ kg/(N s)} = 0.0792 \text{ kg N}^{-1}\,\text{h}^{-1}.$$

All values at 11 km were calculated to be positive, with no negative pressures or temperatures involved, and the engines produce positive thrust at this altitude. Both nozzles are choked and the low pressure turbine exit pressure stays well above ambient, so the turbines have work left over to drive the hot jet. We can also check that the intake can actually carry this mass flow, using the choked mass flow derived in Section 3.3. At 11 km the intake needs to carry 95.8%

of its choked flow, so the flow at the intake face stays subsonic and the part (a) results are physically feasible.

2.2. Results

To evaluate the thrust and TSFC at every altitude, we can pass a basic array of ISA parameters through the equations and produce a graph. No interpolation was used for this section, so resulting values are given in increments of 500 m consistent with the provided ISA table. This was chosen so as not to assume a shape between points without evidence.

Figure 4: Net thrust against altitude at Mach 0.75 and a fuel-air ratio of 3%.

Figure 5: Thrust specific fuel consumption against altitude at Mach 0.75 and a fuel-air ratio of 3%.

Figure 4 reveals a steady decline in thrust from sea level to 15 km (348 kN to 50.2 kN per engine). This aligns directly with the fall in air mass flow into the intake and is an expected result. Both hot and cold nozzles are choked at every altitude, and the specific thrust changes by only approximately 5%, indicating a strong relationship between thrust and mass flow.

TSFC improves slightly with altitude (Figure 5), falling from 0.0833 $\mathrm{kg\,N^{-1}\,h^{-1}}$ at sea level to 0.0792 $\mathrm{kg\,N^{-1}\,h^{-1}}$ at 11 km. The amount of fuel burned per kilogram of core air is already fixed

by the fuel-air ratio, so the improvement is due to the rise in specific thrust as the temperature falls. Above 11 km, the ambient temperature is constant, so TSFC sits constant.

3. Part (b): Maximum Mach number

3.1. Procedure

The cycle, station numbering and assumptions are the same as in part (a). The fuel-air ratio is reduced to 2.5% and the flight Mach number is now variable. The condition that intake conditions may not exceed Mach 1 is applied as an upper limit of 1 on the flight Mach number, so no shock forms ahead of the intake and the subsonic intake relation from part (a) holds at every step. I did include a condition in my code to apply a normal shock at the intake during my testing, however this is outside the scope of the assignment.

At each altitude the full cycle is evaluated at every Mach number step, and the net thrust is checked for sign. The maximum Mach number is the highest step at which the net thrust is still positive. Changing the Mach number changes the thrust in two ways. If raised, the ram drag $\dot{m}C_a$ can grow to equal the gross thrust, or the turbines can fail to supply the fan and compressor work, which shows as the stagnation pressure leaving the low pressure turbine falling to ambient. Both are checked at every step. In the exact same way as before, nozzle choking is checked at every step and altitude, with the unchoked relations used if the critical pressure falls below ambient.

For example, we can check at sea level with a Mach number of 1, where $T_a = 288.15$ K, $p_a = 101.33$ kPa, $\rho_a = 1.2250$ kg/m^3 and $a = 340.29$ m/s. The intake and mass flows follow from the part (a) relations,

$$C_a = 1.0 \cdot 340.29 = 340.29 \text{ m/s}, \quad T_{01} = 288.15 + \frac{340.29^2}{2 \cdot 1005} = 345.76 \text{ K},$$

$$p_{01} = 101.33 \left[1 + \frac{0.95 \cdot 340.29^2}{2 \cdot 1005 \cdot 288.15} \right]^{\frac{1.4}{1.4-1}} = 186.23 \text{ kPa},$$

$$\dot{m} = 1.2250 \cdot 340.29 \cdot 5.067 = 2112.26 \text{ kg/s},$$

$$\dot{m}_c = 358.01 \text{ kg/s}, \quad \dot{m}_b = 1754.25 \text{ kg/s}, \quad \dot{m}_f = 0.025 \cdot 358.01 = 8.9503 \text{ kg/s}.$$

The remaining stagnation states follow from the same relations as in part (a), and are given in Table 1.

Station	01	02	03	04	05	06	07
p_0 (kPa)	186.23	325.91	651.81	5959.43	5840.25	1419.29	204.06
T_0 (K)	345.76	416.29	523.55	1066.75	1851.96	1383.34	931.81

Table 1: Stagnation states at sea level, Mach 1 and a fuel-air ratio of 2.5%.

Since $p_{07} = 204.06$ kPa is greater than $p_a = 101.33$ kPa, the turbines supply the fan and compressor work with pressure left over to drive the hot jet. Checking both nozzles for choking,

$$p_{8,c} = 204.06 \left[1 - \frac{1}{0.95} \left(\frac{1.333 - 1}{1.333 + 1} \right) \right]^{\frac{1.333}{1.333-1}} = 106.34 \text{ kPa},$$

$$p_{9,c} = 325.91\left[1 - \frac{1}{0.95}\left(\frac{1.4-1}{1.4+1}\right)\right]^{\frac{1.4}{1.4-1}} = 165.91 \text{ kPa},$$

which are both greater than $p_a = 101.33$ kPa, so both nozzles are choked, giving $C_8 = 552.81$ m/s, $A_8 = 1.4310$ m^2, $C_9 = 373.35$ m/s and $A_9 = 2.8197$ m^2. The gross thrust is then

$$\begin{aligned}\dot{m}_c(1+f)C_8 + \dot{m}_b C_9 + (p_8 - p_a)A_8 + (p_9 - p_a)A_9 &= 202.86 + 654.94 + 7.18 + 182.11 \\ &= 1047.10 \text{ kN},\end{aligned}$$

which is greater than the ram drag,

$$\dot{m}C_a = 2112.26 \cdot 340.29 = 718.79 \text{ kN},$$

so the net thrust of each engine is $F = 1047.10 - 718.79 = 328$ kN, or 657 kN for both engines. The thrust specific fuel consumption is then

$$\text{TSFC} = \frac{8.9503}{328.31 \times 10^3} = 2.73 \times 10^{-5} \text{ kg/(N s)} = 0.0981 \text{ kg N}^{-1} \text{ h}^{-1}.$$

3.2. Results

Figure 6: Maximum Mach number for positive net thrust against altitude.

Figure 7: Net thrust against altitude at the maximum Mach number.

Figure 8: Thrust specific fuel consumption against altitude at the maximum Mach number.

Figure 6 shows that the maximum Mach number is 1 at every altitude from sea level up to 15 km. We have limited the engines to 1 since the brief states that intake conditions may not exceed Mach 1 [1], and the engine still produces positive net thrust at that limit at every altitude.

To build these figures, the Mach number was stepped from 0.01 to 1 in increments of 0.001, evaluating the full cycle at each step. The maximum is then the last step which has positive thrust, and in our case the thrust is positive at all altitudes and at all Mach numbers. No interpolation has been applied. Additionally, at Mach 1, both nozzles are choked at every altitude, so after supplying work to the compressors and fan, the turbines still have leftover work to drive the jet. For this not to be the case, the fuel-air ratio would need to be smaller. At sea level and Mach 1 this happens below 2.0%, since at 1.9% the low pressure turbine exit pressure falls to 0.975 times ambient and the core flow can no longer leave through the hot nozzle.

The net thrust at Mach 1 falls from 328 kN per engine at sea level to 52.6 kN at 15 km (Figure 7), again following the fall in air mass flow. TSFC falls from 0.0981 $\mathrm{kg\,N^{-1}\,h^{-1}}$ at sea level to 0.0840 $\mathrm{kg\,N^{-1}\,h^{-1}}$ at 11 km and stays constant above 11 km (Figure 8). TSFC is higher

than in part (a) even though the fuel-air ratio is lower, since the larger ram drag at Mach 1 lowers the specific thrust to between 155 and 182 N s/kg, compared to between 220 and 231 N s/kg in part (a).

3.3. Physical feasibility

The cycle needs to be checked for non-physical pressures, temperatures and velocities [3]. At Mach 1, every pressure and temperature in the cycle is positive, the low pressure turbine exit pressure stays above ambient, and both jets leave faster than the flight speed at every altitude.

However, the intake can only carry so much air. The most air the intake face can carry is the choked mass flow,

$$\dot{m}^* = A p_{01} \sqrt{\frac{\gamma_a}{R T_{01}}} \left(\frac{2}{\gamma_a + 1} \right)^{\frac{\gamma_a + 1}{2(\gamma_a - 1)}}. \tag{23}$$

At sea level and Mach 1 this gives 2051.18 kg/s, which is less than the 2112.26 kg/s which the intake requires in our calculations, so in reality the extra air would spill around the nacelle. Repeating this check at every step shows that the intake face chokes at a flight Mach number of about 0.845 at every altitude, so a real engine would be limited by its intake well before Mach 1 under these conditions.

We have also made a set of simplifying assumptions, and more realistic assumptions would lower it further [3]:

- The intake captures a stream tube equal to its own area at every flight speed.
- The fan and compressor pressure ratios and efficiencies are fixed, rather than changing with flight speed.
- There is no limit on the turbine entry temperature, which reaches 1852 K at sea level.

4. Conclusion

At Mach 0.75 and a fuel-air ratio of 3% the net thrust of both engines falls steadily between sea level and a 15 km altitude. Then, with a fuel-air ratio of 2.5%, we can see that the engine in fact produces positive thrust at all altitudes and at all Mach numbers between 0.01 and 1. However, checking the cycle shows that the intake face would choke at a flight Mach number of about 0.85, so the Mach 1 result is only possible under the simplified assumptions used. More realistic modelling of the intake, the turbine entry temperature limit, and the variable compressor pressure ratios and efficiencies would lower the maximum Mach number further.

5. Statement of AI use

Generative AI tool used: Claude Opus 5.5 (Anthropic), through Claude Code.

The tool was used to format the data outputted by the code appropriately, mainly assisting in building descriptive print statements and professional plots. The tool also helped format this report including translating my written equations into Typst format for professionality. No values or figures in this report or in the code are direct AI output. The tool also assisted in pointing out inconsistencies in my draft report, ensuring what I have written satisfies the marking guide and assignment brief. No text in this report was produced by AI.

References

[1] D. Duke, "MAE3405 Assignment 3 – Gas Turbine Design." Monash University, Clayton, VIC, Australia, 2026.

[2] Department of Mechanical and Aerospace Engineering, "MAE3405 Data Tables and Aide Memoire." Monash University, Clayton, VIC, Australia, 2026.

[3] D. Duke, "Assignment 3 Part B." MAE3405 EdStem discussion board, Monash University, Oct. 2026.

6. Appendix

6.1. Python code

All calculations and figures in this report were produced by the following script, `main.py`.

```python
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import scienceplots

FIG_WIDTH = 4.40  # in, 70% of the report text width, one plot per row

plt.style.use("science")
plt.rcParams["figure.figsize"] = (FIG_WIDTH, 2.75)

# -----
# CONVERSION CONSTANTS
# -----

IN_TO_M = 0.0254

# -----
# PARAMETERS - SI UNITS
# Fuel type: Jet-A
# -----

ENGINE_COUNT = 2
BYPASS_RATIO = 4.9
D_INTAKE = 100 * IN_TO_M  # m
C_P_COMP = 1005  # J/kg-K
GAMMA_COMP = 1.4
C_P_TURB = 1148  # J/kg-K
GAMMA_TURB = 1.333
R_EXHAUST = 287  # J/kg-K
T_REF = 298  # K, ref temp
LHV = 43.1e6  # J/kg

PRESSURE_RATIO_OVERALL = 32  # Overall pressure ratio: 32:1
FAN_PRESSURE_RATIO = 1.75
LOW_PRES_COMP_RATIO = 2.0
EFF_C = 0.98  # Combustion efficiency
EFF_ISEN_INTAKE = 0.95  # Intake isentropic efficiency
EFF_ISEN_NOZZLE = 0.95  # Nozzle isentropic efficiency
EFF_MECH_SPOOL = 0.99  # Spool mechanical efficiency
EFF_ISEN = 0.85  # Compressor, fan, and turbines isentropic efficiency
PRESSURE_DROP_COMBUSTOR = 0.02  # 2%

# Import the standard atmosphere table - direct from MAE3405 Aide Memoire
data = pd.read_csv("atmosphere.csv")

# Ambient conditions at each tabulated altitude (0.5 km steps, no interpolation)
altitude = data["altitude_km"].to_numpy() * 1000  # m
Ta = data["temperature_K"].to_numpy()  # K
Pa = data["pressure_Pa"].to_numpy()  # Pa
rho_a = data["density_kg_m3"].to_numpy()  # kg/m^3
sound_speed_a = data["speed_of_sound_m_s"].to_numpy()  # m/s


def engine_performance(M=0.75, FUEL_AIR_RATIO=0.03, sample=None):

    # Flight speed
    C_a = M * sound_speed_a  # m/s

    # Mass flows (useful later)
    MASS_FLOW = rho_a * C_a * np.pi * D_INTAKE**2 * (1 / 4)
    MASS_FLOW_CORE = MASS_FLOW / (1 + BYPASS_RATIO)
    MASS_FLOW_BYPASS = BYPASS_RATIO * MASS_FLOW / (1 + BYPASS_RATIO)
    MASS_FLOW_FUEL = FUEL_AIR_RATIO * MASS_FLOW_CORE

    # Normal shock equations - only applied for M > 1

    M_shock = np.maximum(M, 1.0)  # keeps the shock equations valid when subsonic
```

```
M_i = np.sqrt(
    (1 + (GAMMA_COMP - 1) / 2 * M_shock**2)
    / (GAMMA_COMP * M_shock**2 - (GAMMA_COMP - 1) / 2)
)
P_i = Pa * (1 + 2 * GAMMA_COMP * (M_shock**2 - 1) / (GAMMA_COMP + 1))

# a -> 01 intake

T01 = Ta + (C_a**2) / (2 * C_P_COMP)

P01_sub = Pa * (1 + EFF_ISEN_INTAKE * (C_a**2) / (2 * C_P_COMP * Ta)) ** (
    GAMMA_COMP / (GAMMA_COMP - 1)
)
P01_sup = P_i * (1 + EFF_ISEN_INTAKE * (GAMMA_COMP - 1) / 2 * M_i**2) ** (
    GAMMA_COMP / (GAMMA_COMP - 1)
)

P01 = np.where(M > 1.0, P01_sup, P01_sub)

# 01 -> 02 fan

P02 = FAN_PRESSURE_RATIO * P01
T02s = T01 * (P02 / P01) ** ((GAMMA_COMP - 1) / GAMMA_COMP)
T02 = T01 + (T02s - T01) / EFF_ISEN

# 02 -> 03 LPC

P03 = LOW_PRES_COMP_RATIO * P02
T03s = T02 * (P03 / P02) ** ((GAMMA_COMP - 1) / GAMMA_COMP)
T03 = T02 + (T03s - T02) / EFF_ISEN

# 03 -> 04 HPC

high_pres_comp_ratio = PRESSURE_RATIO_OVERALL / (
    FAN_PRESSURE_RATIO * LOW_PRES_COMP_RATIO
)
P04 = high_pres_comp_ratio * P03
T04s = T03 * (P04 / P03) ** ((GAMMA_COMP - 1) / GAMMA_COMP)
T04 = T03 + (T04s - T03) / EFF_ISEN

# 04 -> 05 combustor

P05 = (1 - PRESSURE_DROP_COMBUSTOR) * P04
T05 = T_REF + (C_P_COMP * (T04 - T_REF) + EFF_C * FUEL_AIR_RATIO * LHV) / (
    (1 + FUEL_AIR_RATIO) * C_P_TURB
)

# 05 -> 06 HPT

T06 = T05 - (C_P_COMP * (T04 - T03)) / (
    EFF_MECH_SPOOL * (1 + FUEL_AIR_RATIO) * C_P_TURB
)
T06s = T05 - (T05 - T06) / EFF_ISEN
P06 = P05 * (T06s / T05) ** (GAMMA_TURB / (GAMMA_TURB - 1))

# 06 -> 07 LPT

T07 = T06 - C_P_COMP * (
    ((T03 - T02) + (1 + BYPASS_RATIO) * (T02 - T01))
    / (EFF_MECH_SPOOL * (1 + FUEL_AIR_RATIO) * C_P_TURB)
)
T07s = T06 - (T06 - T07) / EFF_ISEN
P07 = P06 * (T07s / T06) ** (GAMMA_TURB / (GAMMA_TURB - 1))

# 07 -> 8 hot nozzle

P8_crit = P07 * (
    1 - (1 / EFF_ISEN_NOZZLE) * ((GAMMA_TURB - 1) / (GAMMA_TURB + 1))
) ** (GAMMA_TURB / (GAMMA_TURB - 1))

choked_8 = P8_crit > Pa  # checked at each altitude

# choked: sonic exit at the critical pressure
T8_choked = 2 * T07 / (GAMMA_TURB + 1)
C8_choked = np.sqrt(GAMMA_TURB * R_EXHAUST * T8_choked)
rho_8 = P8_crit / (R_EXHAUST * T8_choked)
A8_choked = MASS_FLOW_CORE * (1 + FUEL_AIR_RATIO) / (rho_8 * C8_choked)
```

```
    # unchoked: full expansion to ambient, no pressure thrust
    T8s = T07 * (Pa / P07) ** ((GAMMA_TURB - 1) / GAMMA_TURB)
    T8_unchoked = T07 - EFF_ISEN_NOZZLE * (T07 - T8s)
    C8_unchoked = np.sqrt(2 * C_P_TURB * (T07 - T8_unchoked))

    P8 = np.where(choked_8, P8_crit, Pa)
    T8 = np.where(choked_8, T8_choked, T8_unchoked)
    C8 = np.where(choked_8, C8_choked, C8_unchoked)
    A8 = np.where(choked_8, A8_choked, 0)

    # 02 -> 9 cold nozzle

    P9_crit = P02 * (
        1 - (1 / EFF_ISEN_NOZZLE) * ((GAMMA_COMP - 1) / (GAMMA_COMP + 1))
    ) ** (GAMMA_COMP / (GAMMA_COMP - 1))

    choked_9 = P9_crit > Pa  # checked at each altitude

    # choked: sonic exit at the critical pressure
    T9_choked = 2 * T02 / (GAMMA_COMP + 1)
    C9_choked = np.sqrt(GAMMA_COMP * R_EXHAUST * T9_choked)
    rho_9 = P9_crit / (R_EXHAUST * T9_choked)
    A9_choked = MASS_FLOW_BYPASS / (rho_9 * C9_choked)

    # unchoked: full expansion to ambient, no pressure thrust
    T9s = T02 * (Pa / P02) ** ((GAMMA_COMP - 1) / GAMMA_COMP)
    T9_unchoked = T02 - EFF_ISEN_NOZZLE * (T02 - T9s)
    C9_unchoked = np.sqrt(2 * C_P_COMP * (T02 - T9_unchoked))

    P9 = np.where(choked_9, P9_crit, Pa)
    T9 = np.where(choked_9, T9_choked, T9_unchoked)
    C9 = np.where(choked_9, C9_choked, C9_unchoked)
    A9 = np.where(choked_9, A9_choked, 0)

    # Net thrust
    F = (
        MASS_FLOW_CORE * (1 + FUEL_AIR_RATIO) * C8
        + MASS_FLOW_BYPASS * C9
        - MASS_FLOW * C_a
        + (P8 - Pa) * A8
        + (P9 - Pa) * A9
    )

    F_net = F * ENGINE_COUNT

    # TSFC

    TSFC = MASS_FLOW_FUEL / F

    # This part just prints out sample values at one altitude for the report
    if sample is not None:
        i = sample
        print(f"""
Sample calculation at {altitude[i] / 1000:.1f} km (M = {np.broadcast_to(M, Ta.shape)[i]:.2f}, f = {FUEL_AIR_RATIO})
Ambient: Ta = {Ta[i]:.2f} K, Pa = {Pa[i] / 1e3:.2f} kPa, rho_a = {rho_a[i]:.4f} kg/m^3, a = {sound_speed_a[i]:.2f} m/
s
Intake area: A = {np.pi * D_INTAKE**2 / 4:.3f} m^2
C_a = {C_a[i]:.2f} m/s
T01 = {T01[i]:.2f} K, P01 = {P01[i] / 1e3:.2f} kPa
m_dot = {MASS_FLOW[i]:.2f} kg/s, m_dot_core = {MASS_FLOW_CORE[i]:.2f} kg/s, m_dot_bypass = {MASS_FLOW_BYPASS[i]:.2f}
kg/s, m_dot_fuel = {MASS_FLOW_FUEL[i]:.4f} kg/s
P02 = {P02[i] / 1e3:.2f} kPa, T02s = {T02s[i]:.2f} K, T02 = {T02[i]:.2f} K
P03 = {P03[i] / 1e3:.2f} kPa, T03s = {T03s[i]:.2f} K, T03 = {T03[i]:.2f} K
HPC ratio = {high_pres_comp_ratio:.4f}
P04 = {P04[i] / 1e3:.2f} kPa, T04s = {T04s[i]:.2f} K, T04 = {T04[i]:.2f} K
P05 = {P05[i] / 1e3:.2f} kPa, T05 = {T05[i]:.2f} K
T06 = {T06[i]:.2f} K, T06s = {T06s[i]:.2f} K, P06 = {P06[i] / 1e3:.2f} kPa
T07 = {T07[i]:.2f} K, T07s = {T07s[i]:.2f} K, P07 = {P07[i] / 1e3:.2f} kPa
P8_crit = {P8_crit[i] / 1e3:.2f} kPa, choked_8 = {choked_8[i]}
T8 = {T8[i]:.2f} K, C8 = {C8[i]:.2f} m/s, rho_8 = {rho_8[i]:.4f} kg/m^3, A8 = {A8[i]:.4f} m^2
P9_crit = {P9_crit[i] / 1e3:.2f} kPa, choked_9 = {choked_9[i]}
T9 = {T9[i]:.2f} K, C9 = {C9[i]:.2f} m/s, rho_9 = {rho_9[i]:.4f} kg/m^3, A9 = {A9[i]:.4f} m^2
Core momentum = {MASS_FLOW_CORE[i] * (1 + FUEL_AIR_RATIO) * C8[i] / 1e3:.2f} kN
Bypass momentum = {MASS_FLOW_BYPASS[i] * C9[i] / 1e3:.2f} kN
Ram drag = {MASS_FLOW[i] * C_a[i] / 1e3:.2f} kN
Hot pressure thrust = {(P8[i] - Pa[i]) * A8[i] / 1e3:.2f} kN
Cold pressure thrust = {(P9[i] - Pa[i]) * A9[i] / 1e3:.2f} kN
Gross thrust (all terms except ram drag) = {(F[i] + MASS_FLOW[i] * C_a[i]) / 1e3:.2f} kN
```

```
F (per engine) = {F[i] / 1e3:.2f} kN, F_net (both engines) = {F_net[i] / 1e3:.2f} kN
TSFC = {TSFC[i]:.4e} kg/(N.s) = {TSFC[i] * 3600:.4f} kg/(N.h)
""")

    # Extra quantities used to support the discussion in the report
    extra = {
        "mass_flow": MASS_FLOW,
        "specific_thrust": F / MASS_FLOW,  # N s/kg, per engine
        "choked_8": choked_8,
        "choked_9": choked_9,
        "P07_ratio": P07 / Pa,
        "T01": T01,
        "P01": P01,
        "T05": T05,
        "C_a": C_a,
        "C8": C8,
        "C9": C9,
    }

    return F_net, TSFC, extra


# -----
# PART A
# -----

# Altitude indices used for the values quoted in the report
i_11km = np.flatnonzero(altitude == 11000)[0]
above_11km = altitude >= 11000

F_net, TSFC, extra_a = engine_performance(sample=i_11km)

# Plots

FIGURES_DIR = Path("figures")
FIGURES_DIR.mkdir(exist_ok=True)

altitude_km = altitude / 1000

fig_thrust, ax_thrust = plt.subplots()
ax_thrust.plot(
    altitude_km,
    (F_net / ENGINE_COUNT) / 1e3,
    marker="o",
    markersize=2,
    label="Per engine",
)
ax_thrust.plot(altitude_km, F_net / 1e3, marker="s", markersize=2, label="Both engines")
ax_thrust.set_xlabel("Altitude (km)")
ax_thrust.set_ylabel("Net thrust (kN)")
ax_thrust.grid(True)
ax_thrust.legend()
fig_thrust.savefig(FIGURES_DIR / "part_a_thrust.pdf")

fig_tsfc, ax_tsfc = plt.subplots()
ax_tsfc.plot(altitude_km, TSFC * 3600, marker="o", markersize=2)
ax_tsfc.set_xlabel("Altitude (km)")
ax_tsfc.set_ylabel(r"TSFC (kg\,N$^{-1}$\,h$^{-1}$)")
ax_tsfc.grid(True)
fig_tsfc.savefig(FIGURES_DIR / "part_a_tsfc.pdf")

spec_a = extra_a["specific_thrust"]

print(f"""
PART A RESULTS (M = 0.75, f = 0.03)
Thrust at sea level: {F_net[0] / 1e3:.2f} kN ({F_net[0] / ENGINE_COUNT / 1e3:.2f} kN per engine)
Thrust at 15km: {F_net[-1] / 1e3:.2f} kN ({F_net[-1] / ENGINE_COUNT / 1e3:.2f} kN per engine)
Air mass flow per engine: {extra_a["mass_flow"][0]:.0f} kg/s at sea level, {extra_a["mass_flow"][-1]:.0f} kg/s at 15
km
Both nozzles choked at every altitude: {np.all(extra_a["choked_8"] & extra_a["choked_9"])}
Specific thrust: {spec_a.min():.0f} to {spec_a.max():.0f} N.s/kg, a change of {(spec_a.max() / spec_a.min() - 1) *
100:.1f}%
TSFC at sea level: {TSFC[0] * 3600:.4f} kg/(N.h)
TSFC at 11 km: {TSFC[i_11km] * 3600:.4f} kg/(N.h)
TSFC at 15 km: {TSFC[-1] * 3600:.4f} kg/(N.h)
TSFC above 11 km: {TSFC[above_11km].min() * 3600:.4f} to {TSFC[above_11km].max() * 3600:.4f} kg/(N.h)
""")
```

```
# -----
# PART B
# -----

"""
Now we may consider different airspeeds at the reduced fuel-air ratio.

The intake of a high-bypass turbofan is not designed for supersonic flow, so the
flight Mach number is limited to 1. The maximum Mach number at each altitude is
the last step with positive net thrust, up to and including Mach 1.

Net thrust and TSFC are then taken at that maximum Mach number.
"""

mach_steps = np.linspace(0.01, 1.0, 991)  # steps of 0.001, limited to Mach 1
results_b = [engine_performance(M, 0.025) for M in mach_steps]
thrust = np.array([F for F, _, _ in results_b])
tsfc = np.array([sfc for _, sfc, _ in results_b])

# Turbine work check: the LPT exit stagnation pressure must stay above ambient
P07_ratio_steps = np.array([extra["P07_ratio"] for _, _, extra in results_b])

# Take the last positive step at each altitude

last = (thrust > 0).sum(axis=0) - 1

M_max = mach_steps[last]

# Net thrust and TSFC at the maximum Mach number

columns = np.arange(len(altitude))
F_net_b = thrust[last, columns]
TSFC_b = tsfc[last, columns]

# Re-evaluate the cycle at the maximum Mach number for the sample calculation at
# sea level and the supporting values in the report
_, _, extra_b = engine_performance(M_max, 0.025, sample=0)

# Plots

fig_mach_b, ax_mach_b = plt.subplots()
ax_mach_b.plot(altitude_km, M_max, marker="o", markersize=2)
ax_mach_b.set_xlabel("Altitude (km)")
ax_mach_b.set_ylabel("Maximum Mach number")
ax_mach_b.set_ylim(0, 1.2)
ax_mach_b.grid(True)
fig_mach_b.savefig(FIGURES_DIR / "part_b_mach.pdf")

fig_thrust_b, ax_thrust_b = plt.subplots()
ax_thrust_b.plot(
    altitude_km,
    (F_net_b / ENGINE_COUNT) / 1e3,
    marker="o",
    markersize=2,
    label="Per engine",
)
ax_thrust_b.plot(
    altitude_km, F_net_b / 1e3, marker="s", markersize=2, label="Both engines"
)
ax_thrust_b.set_xlabel("Altitude (km)")
ax_thrust_b.set_ylabel("Net thrust (kN)")
ax_thrust_b.grid(True)
ax_thrust_b.legend()
fig_thrust_b.savefig(FIGURES_DIR / "part_b_thrust.pdf")

fig_tsfc_b, ax_tsfc_b = plt.subplots()
ax_tsfc_b.plot(altitude_km, TSFC_b * 3600, marker="o", markersize=2)
ax_tsfc_b.set_xlabel("Altitude (km)")
ax_tsfc_b.set_ylabel(r"TSFC (kg\,N$^{-1}$\,h$^{-1}$)")
ax_tsfc_b.grid(True)
fig_tsfc_b.savefig(FIGURES_DIR / "part_b_tsfc.pdf")

spec_b = extra_b["specific_thrust"]
P07_ratio_b = extra_b["P07_ratio"]

print(f"""
PART B RESULTS (f = 0.025, M <= 1)
Net thrust positive at every Mach step and altitude: {np.all(thrust > 0)}
```

```
LPT exit stagnation pressure above ambient at every Mach step and altitude: {np.all(P07_ratio_steps > 1)}
Maximum Mach number: {M_max.min():.3f} to {M_max.max():.3f}
Both nozzles choked at the maximum Mach number at every altitude: {np.all(extra_b["choked_8"] & extra_b["choked_9"])}
P07/Pa at the maximum Mach number: {P07_ratio_b[0]:.2f} at sea level, {P07_ratio_b[above_11km].min():.2f} to
{P07_ratio_b[above_11km].max():.2f} above 11 km
Thrust at sea level: {F_net_b[0] / 1e3:.2f} kN ({F_net_b[0] / ENGINE_COUNT / 1e3:.2f} kN per engine)
Thrust at 15km: {F_net_b[-1] / 1e3:.2f} kN ({F_net_b[-1] / ENGINE_COUNT / 1e3:.2f} kN per engine)
Air mass flow per engine: {extra_b["mass_flow"][0]:.0f} kg/s at sea level, {extra_b["mass_flow"][-1]:.0f} kg/s at 15
km
Specific thrust: {spec_b.min():.0f} to {spec_b.max():.0f} N.s/kg, smallest at {altitude_km[spec_b.argmin()]:.1f} km
TSFC at sea level: {TSFC_b[0] * 3600:.4f} kg/(N.h)
TSFC at 11 km: {TSFC_b[i_11km] * 3600:.4f} kg/(N.h)
TSFC at 15 km: {TSFC_b[-1] * 3600:.4f} kg/(N.h)
TSFC above 11 km: {TSFC_b[above_11km].min() * 3600:.4f} to {TSFC_b[above_11km].max() * 3600:.4f} kg/(N.h)
""")

# -----
# PHYSICAL FEASIBILITY
# -----

"""
The intake face has the same area as the intake, so all of the captured air must
pass through it at the stagnation state 01. The most it can carry is the choked
mass flow, when the flow at the intake face reaches Mach 1.
"""

INTAKE_AREA = np.pi * D_INTAKE**2 / 4  # m^2


def choked_mass_flow(P0, T0):
    return (
        INTAKE_AREA
        * P0
        * np.sqrt(GAMMA_COMP / (R_EXHAUST * T0))
        * (2 / (GAMMA_COMP + 1)) ** ((GAMMA_COMP + 1) / (2 * (GAMMA_COMP - 1)))
    )


intake_ratio_a = extra_a["mass_flow"] / choked_mass_flow(extra_a["P01"], extra_a["T01"])
intake_ratio_steps = np.array(
    [
        extra["mass_flow"] / choked_mass_flow(extra["P01"], extra["T01"])
        for _, _, extra in results_b
    ]
)

# First Mach step at which the intake face would need to pass more than the choked flow
M_intake_choke = mach_steps[(intake_ratio_steps < 1).sum(axis=0)]

# Lowest fuel-air ratio that still gives positive thrust at Mach 1 at every altitude.
# Below this the LPT exit pressure falls under ambient and the jet velocity is undefined.
fuel_air_steps = np.round(np.arange(0.025, 0.0095, -0.001), 3)
with np.errstate(invalid="ignore"):
    results_f = [engine_performance(1.0, f) for f in fuel_air_steps]
P07_ratio_f = np.array([extra["P07_ratio"].min() for _, _, extra in results_f])
positive_at_mach_1 = np.array([np.all(F > 0) for F, _, _ in results_f])
j_f = np.flatnonzero(positive_at_mach_1)[-1]  # lowest working fuel-air ratio

print(f"""
PHYSICAL FEASIBILITY
Part A m_dot / choked intake flow at 11 km: {intake_ratio_a[i_11km]:.3f}
Sea level, Mach 1: m_dot = {extra_b["mass_flow"][0]:.2f} kg/s, choked intake flow = {choked_mass_flow(extra_b["P01"]
[0], extra_b["T01"][0]):.2f} kg/s
Flight Mach at which the intake face chokes: {M_intake_choke.min():.3f} to {M_intake_choke.max():.3f}
Part B maximum T05: {extra_b["T05"].max():.0f} K
Both jets faster than the flight speed at Mach 1 at every altitude: {np.all((extra_b["C8"] > extra_b["C_a"]) &
(extra_b["C9"] > extra_b["C_a"]))}
Lowest fuel-air ratio with positive thrust at Mach 1 at every altitude: {fuel_air_steps[j_f]:.3f} (min P07/Pa =
{P07_ratio_f[j_f]:.3f})
At f = {fuel_air_steps[j_f + 1]:.3f}: min P07/Pa = {P07_ratio_f[j_f + 1]:.3f}
""")

# -----
# SUPERSONIC SWEEP - supporting figures beyond the Mach 1 limit of part B
# -----

"""
Here the Mach 1 limit is lifted to see where the thrust actually reaches zero.
```

```
We must consider the possibility that the ambient airspeed is supersonic and that
the presence of a normal shock at the intake will lower the intake to subsonic.

Let's use normal shock theory and define a new state, the intake state (i), which sits
between (a) and (01) just inside the shock at the start of the intake nozzle.

NOTE FOR CONSISTENCY:
The aide memoire lists normal shock tables for hand calculations.
I have chosen to use the normal shock theory equations (which are what the table uses)
since this is better for code readability and accuracy.
"""

mach_sweep = np.round(np.arange(0.01, 2.80, 0.001), 3)
results_sweep = [engine_performance(M, 0.025) for M in mach_sweep]
thrust_sweep = np.array([F for F, _, _ in results_sweep])
tsfc_sweep = np.array([sfc for _, sfc, _ in results_sweep])

# Take the last positive step at each altitude

M_zero_thrust = mach_sweep[(thrust_sweep > 0).sum(axis=0) - 1]

# Thrust and TSFC against Mach number at selected altitudes, up to the last
# step with positive thrust. The dashed line marks Mach 1, where the shock forms.

FIG_WIDTH_FULL = 6.30  # in, full report text width
SUBFIGURE_SIZE = (FIG_WIDTH_FULL, FIG_WIDTH_FULL * 2.625 / 3.5)
PLOT_ALTITUDES_KM = (0, 5, 10, 15)

fig_thrust_sweep, axes_thrust_sweep = plt.subplots(
    2, 2, sharex=True, figsize=SUBFIGURE_SIZE, layout="constrained"
)
for ax, km in zip(axes_thrust_sweep.flat, PLOT_ALTITUDES_KM):
    j = np.argmin(np.abs(altitude_km - km))
    positive = thrust_sweep[:, j] > 0
    ax.plot(
        mach_sweep[positive],
        (thrust_sweep[positive, j] / ENGINE_COUNT) / 1e3,
        label="Per engine",
    )
    ax.plot(mach_sweep[positive], thrust_sweep[positive, j] / 1e3, label="Both engines")
    ax.axvline(1.0, color="r", linestyle="--", linewidth=1.0, label="Mach 1")
    ax.set_title(f"{km} km")
    ax.grid(True)
for ax in axes_thrust_sweep[-1, :]:
    ax.set_xlabel("Flight Mach number")
for ax in axes_thrust_sweep[:, 0]:
    ax.set_ylabel("Net thrust (kN)")
fig_thrust_sweep.legend(
    *axes_thrust_sweep[0, 0].get_legend_handles_labels(),
    loc="outside lower center",
    ncols=3,
)
fig_thrust_sweep.savefig(FIGURES_DIR / "sweep_thrust.pdf")

fig_tsfc_sweep, axes_tsfc_sweep = plt.subplots(
    2, 2, sharex=True, sharey=True, figsize=SUBFIGURE_SIZE, layout="constrained"
)
for ax, km in zip(axes_tsfc_sweep.flat, PLOT_ALTITUDES_KM):
    j = np.argmin(np.abs(altitude_km - km))
    positive = thrust_sweep[:, j] > 0
    ax.plot(mach_sweep[positive], tsfc_sweep[positive, j] * 3600)
    ax.axvline(1.0, color="r", linestyle="--", linewidth=1.0, label="Mach 1")
    ax.set_yscale("log")
    ax.set_title(f"{km} km")
    ax.grid(True)
for ax in axes_tsfc_sweep[-1, :]:
    ax.set_xlabel("Flight Mach number")
for ax in axes_tsfc_sweep[:, 0]:
    ax.set_ylabel(r"TSFC (kg\,N$^{-1}$\,h$^{-1}$)")
fig_tsfc_sweep.legend(
    *axes_tsfc_sweep[0, 0].get_legend_handles_labels(),
    loc="outside lower center",
    ncols=3,
)
fig_tsfc_sweep.savefig(FIGURES_DIR / "sweep_tsfc.pdf")

print(f"""
```

```
SUPERSONIC SWEEP (f = 0.025, normal shock above Mach 1)
Zero-thrust Mach number at sea level: {M_zero_thrust[0]:.3f}
Zero-thrust Mach number above 11 km: {M_zero_thrust[above_11km].min():.3f} to {M_zero_thrust[above_11km].max():.3f}
Zero-thrust Mach number at 15 km: {M_zero_thrust[-1]:.3f}
""")
```